

Patrick Salsbury

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EXPERIENCE

Data Engineer | Tesla - Palo Alto, CA

January 2025 - Present

- Eliminated 20+ hours of manual engineering work per week by designing and owning a production-grade **Dagster** orchestration platform on **Kubernetes**, delivering 30+ automated pipelines spanning ETL, anomaly detection, real-time alerting, and other jobs across the hardware test org.
- Reduced time-series query latency by 20x over a legacy Apache **Iceberg** architecture by leading the design and migration to a **ClickHouse** database housing 100B+ rows, directly accelerating vehicle part data analysis turnaround for test and design engineering teams.
- Cut end-to-end latency from 30 seconds to under 500ms by engineering a high-throughput **PyFlink/Kafka** streaming pipeline capable of processing 3M+ messages per minute across 100+ hardware test devices, enabling real-time tester and test component observability at scale
- Centralized component traceability for 50k+ parts across 100+ test devices by designing a **MySQL** metadata database and a **FastAPI** REST API, replacing manual tracking workflows and improving data accuracy and availability for vehicle reliability testing programs
- Processed petabytes of vehicle test data by building and maintaining 20+ production batch pipelines in **Airflow** and **Spark**, integrating various sources including **Iceberg**, **MySQL**, **PostgreSQL**, and other sources to power cross-functional dashboards and stakeholder reporting
- Eliminated manual deployment to **Linux** VMs by migrating services (Dagster, FastAPI, Flink Pipeline) to a production **Kubernetes** cluster with full CI/CD using **Github actions**.
- Increased reliability of data systems and platforms by implementing observability across all applications by **Prometheus** to emit health check metrics and **Grafana** to visualize system behavior in real-time.

Data Science Intern | Tesla - Palo Alto, CA

June 2024 - December 2024

- Developed a comprehensive internal data analytics application for the hardware test team, enabling device usage monitoring, KPI tracking, and other statistical analysis, all by using **Plotly Dash** hosted on **Kubernetes**.
- Deployed an Isolation Forest **Anomaly Detection Model** to automatically identify manufacturing defects and supplier variability, further increasing the reliability of oil pumps by 50%, fostering population-wide analysis.
- Analyzed over 10 billion entries of fleet data to identify root causes of vehicle fault errors; went forward and created an interactive dashboard using **Bokeh Panel** and **Trino** to showcase these different key performance indicators and allow users to identify issues themselves.

Data Engineer Intern | EverCharge - Palo Alto, CA

June 2023 - September 2023

- Identified bugs, hardware issues, and installation problems that were previously unknown affecting 5-10% of all products by performing time series analysis, inferential analysis, and anomaly detection using **Python** and **SQL**.
- Engineered **Retool** dashboards to showcase **KPIs** regarding device performance while performing **PostgreSQL** query optimization.
- Published 3 different data science reports that communicated answers to stakeholders' questions and highlighted areas of concern.
- Initiated a proof-of-concept project that involved training an ML anomaly detection model to identify abnormal device behavior.

Data Science Intern | HM Electronics - Carlsbad, CA

June 2022 - September 2022

- Analyzed customers' device telemetry data, improving production quality and customer relationships by showcasing areas of concern.
- Extracted semi-structured console log data from 100+ customer devices to ingest into a **Azure Blob Data Lake**.
- Automated the **ETL** process on 30+ million raw records of data using **PySpark** on **Databricks(Microsoft Azure)**.
- Predicted customer product device failure using **Classification** and **Clustering ML** models, isolating variable indicators for tech support.
- Led **Agile** sprint presentations to communicate findings using **Python**, **SQL**, and **Power BI** to influence decision-making of product managers and software engineers.

PROJECTS

Real-Time Weather Data Dashboard Pipeline

- Constructed dynamic **dashboards** visualizing real-time weather updates using **Tableau**, **MongoDB**, **Docker**, and weather APIs.
- Highlighted temperature, humidity, and other attributes using **Apache Airflow** to build an ETL **batch** processing pipeline that ingested, transformed, and loaded the data while performing various quality checks.

E-commerce Recommender System

- Developed a machine learning model for an e-commerce recommender system using **Sci-kit Learn**, **H2o AutoML**, and **Tensorflow**.
- Performed exploratory data analysis, feature engineering, model training and validating, and hyperparameter tuning in order to obtain a model with an accuracy, F1-score, and ROC-AUC of 0.71, 0.72, 0.79 respectively.
- Deployed the finalized Collaborative Filtering Tensorflow model using **Flask** and **HTML** to emulate a real-world recommender system user experience.

EDUCATION

Columbia University | M.S. Computer Science - *Distributed Systems Specialization* (Remote, Part-time)

January 2026 - Present

University of California, San Diego | B.S. Data Science

September 2021 - March 2025

TECHNICAL SKILLS

Languages: Python, SQL, Java

Databases: Iceberg/S3, ClickHouse, MySQL PostgreSQL

Libraries / Frameworks: PySpark, Pandas, Polars, FastAPI, Plotly, Dash, SciKit-Learn, SciPy, Dash, XGBoost, TensorFlow, PyTorch

Softwares: Dagster, Airflow, Spark, Flink, Kafka, Grafana, Git, Docker, Kubernetes, Power BI, Tableau, Retool